

Problem 4.4

What principal will earn interest of 100 in 7 years at a simple interest rate of 6%?

Solution

For simple interest,

$$A(t) = P(1 + it),$$

and the interest earned is

$$I = P \cdot i \cdot t.$$

Here,

$$t = 7, \quad i = 6\% = 0.06, \quad I = 100.$$

So,

$$100 = P(0.06)(7).$$

Therefore,

$$P = \frac{100}{0.42}.$$

Hence,

$$P \approx 238.095.$$